Piping Connection Symbol Legend

Symbols that show how two pipe ends are joined. A connection symbol sits at a point where pipe segments meet or terminate, and the type is identified by the shape of the mark at that point.

All pipe lines are drawn black. Line weight and line color carry no meaning — only the mark at the junction does. Each symbol is shown three ways: inline between two straight lengths, at an elbow, and at a tee.

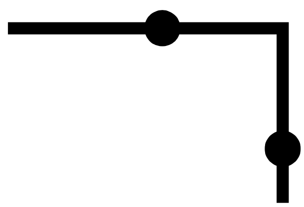
Two determinations are made at each connection. First, the **connection type**, from the symbol shape. Second, **ShopField**, recorded as **1** for a shop weld or **2** for a field weld, from whether the mark carries an X or rays.

1. Butt Weld — plain dot

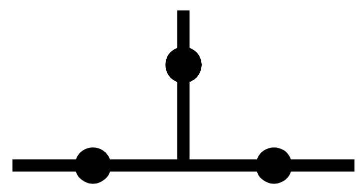
Identify by: a single solid filled dot centered on the pipe line, with nothing radiating from it and nothing crossing it.

Connection type is **butt weld**. ShopField is **1** (shop).

Inline



At an elbow



At a tee

2. Field Weld — dot with an X or rays

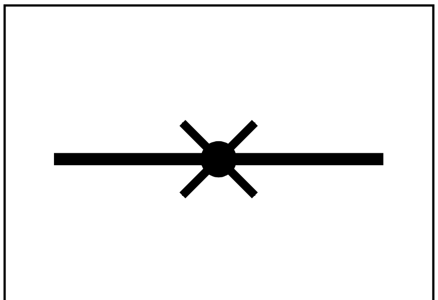
Identify by: a solid filled dot with an X drawn through it, or with short strokes radiating from it. The full X passing straight through the dot is the most common form; the rays are a variant of the same marking. Both mean the same thing.

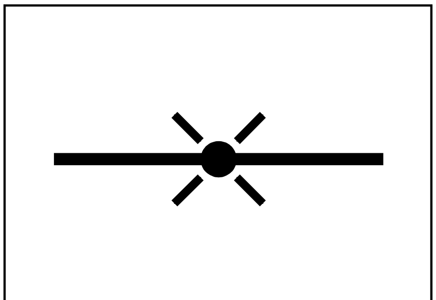
This is still a butt weld. The connection type is **butt weld**, identical to section 1. The X or rays do not change the connection type — they record only where the weld is performed. Do not treat butt weld and field weld as competing choices.

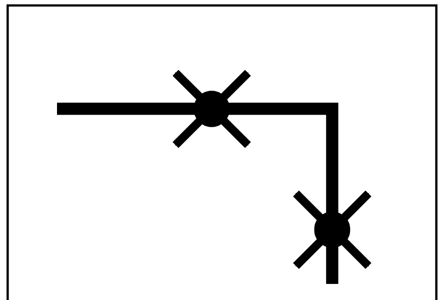
ShopField is **2** (field), meaning the weld is not performed in the shop but on site during installation. Whenever this marking is found, ShopField is 2 without exception.

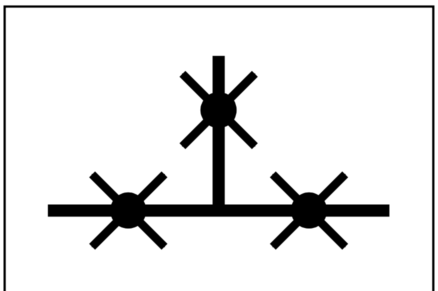
The X or rays are sometimes absent. A field weld is often indicated instead by the letters **FW** written near the weld, or on a leader line pointing at it. This annotation carries the same meaning as the X: connection type **butt weld**, ShopField **2.**

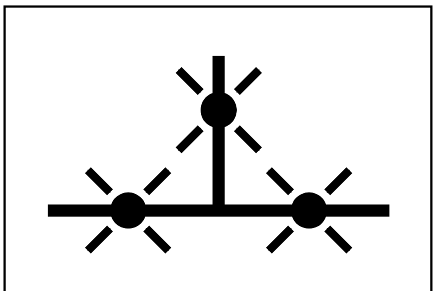
An FW annotation may be part of the original drawing, in the same color as everything else, or it may be a markup added later in a different color — **most often red.** Treat a colored FW markup as equally authoritative. Read it whether it is printed in black or added in color, and associate it with the nearest weld mark it sits beside or points to.

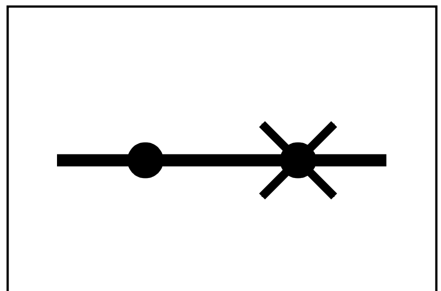

Inline — full X (most common)

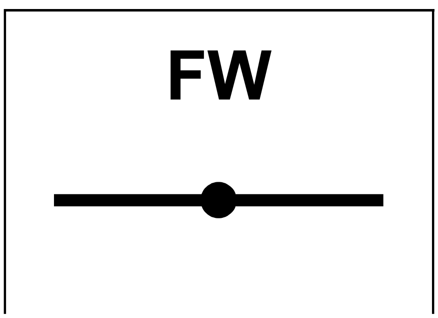

Inline — rays variant

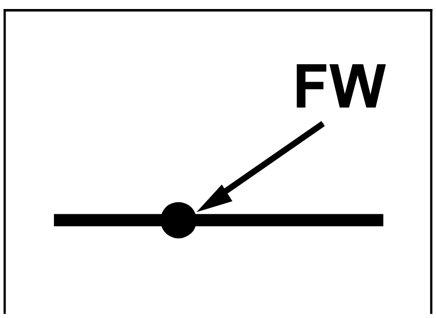

At an elbow — full X

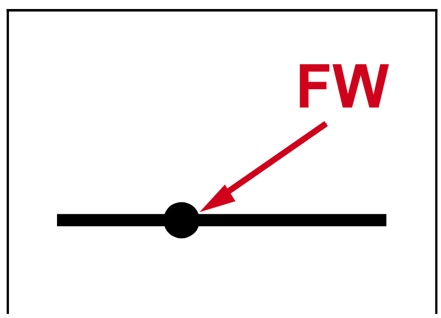

At a tee — full X


At a tee — rays variant


Both on one run — left ShopField 1, right ShopField 2







FW written beside the weld — no X

FW on a leader pointing at the weld

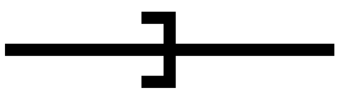
FW added later as a red markup — same meaning

3. Socket Weld or Threaded — bracket

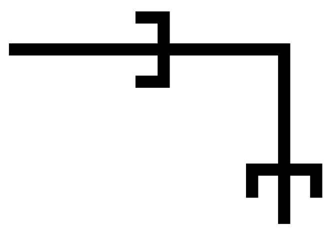
Identify by: a bracket, drawn as a bar across the pipe with two short stubs turning off its ends toward the incoming pipe, forming a shallow cup or open square around the joint. No filled dot is present.

This symbol is ambiguous by itself. It represents a socket weld, but it is very often used to represent a threaded connection as well. The symbol alone cannot settle which one it is.

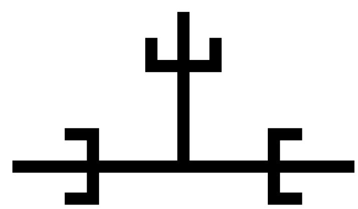
Resolve from the bill of materials. Find the fitting that carries this symbol in the BOM and read its definition — the fitting description states whether it is socket weld or threaded. Do not decide from the symbol alone. ShopField is determined separately, by the presence of an X or rays.



Inline



At an elbow

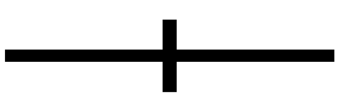


At a tee

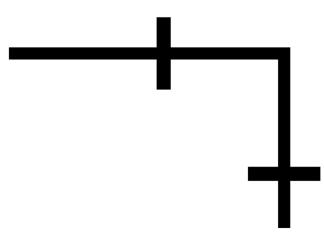
4. Threaded Connection — plain perpendicular bar

Identify by: a single straight bar crossing the pipe at a right angle, with no dot and no stubs turning off its ends. The plain perpendicular bar is the whole symbol. Sometimes annotated *THD* or *SCR* (screwed).

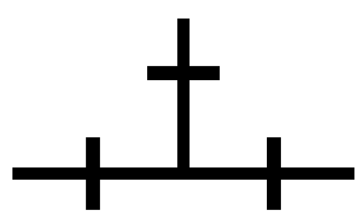
Interchangeable with the bracket in section 3. A threaded connection may be drawn either as this bar or as the socket weld bracket, and drafters frequently use the bracket for threaded connections. Finding this bar suggests threaded, but the bill of materials remains the deciding source: look up the fitting and read its definition before recording the type.



Inline



At an elbow



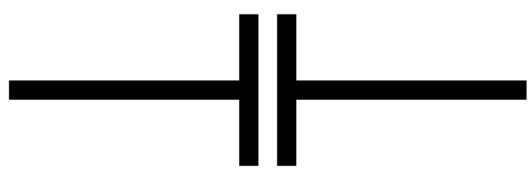
At a tee

5. Flanged Connection — bolt-up

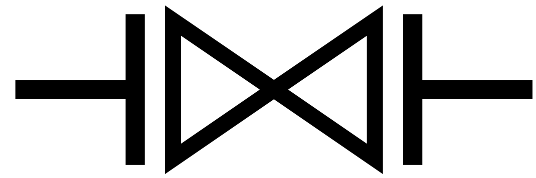
Identify by: a bar across the pipe representing the flange face, paired with a second bar facing it across a small gap. Two facing bars with a gap between them is one **bolt-up**. A flange bar is longer than a threaded bar and always sits at the end of a pipe segment or at the face of a component.

A valve, blind, or other flanged component drawn between two flange faces produces **one bolt-up on each side** — count both. A line-ending flange meeting a blind flange is one bolt-up.

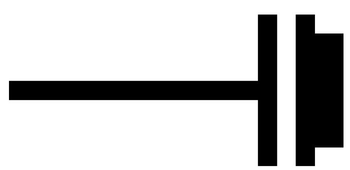
Half bolt-up. If the component on the other side of the flange is drawn in dashed lines, the bolt-up does not occur within the scope of this drawing — the dashed item is future, existing, or by others. Record it as a **half bolt-up**, not a full one. Any flange face without a complete mating face shown in solid lines is a half bolt-up.



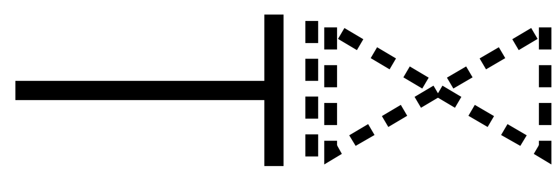
Flange to flange — one bolt-up



Flanged valve — two bolt-ups, one each side



Line-ending flange to a blind — one bolt-up



Flange facing a dashed component — half bolt-up

How to tell them apart

Apply these tests in order at each junction. Connection type and ShopField are two separate answers.

1. Is there a solid filled dot? If yes, connection type is **butt weld**. Then check the dot for an X through it or rays around it: present means ShopField **2**, absent means ShopField **1**.
2. If the dot is plain, check for an **FW** annotation beside it or on a leader pointing at it, in any color including red markup. Found means ShopField **2**. Only a plain dot with no X, no rays, and no FW nearby is ShopField **1.**
3. No dot, but a bar crossing the pipe? Look at the ends of that bar. Stubs turning off both ends is the **bracket** (section 3); plain square ends is the **bar** (section 4). Either way, read the fitting definition in the bill of materials to record socket weld or threaded.
4. Is there a long bar at the end of a pipe segment or at the face of a component, facing another bar across a small gap? That is a **flanged bolt-up**. Count one per facing pair — a component between two flange faces is two. If the mating side is drawn dashed rather than solid, record a **half bolt-up**.
5. Neither a dot nor a perpendicular bar means this is not one of the connection types above. Do not force a match — report it as unidentified.

The same symbol keeps its meaning at any orientation. A mark on a vertical pipe reads the same as on a horizontal one, and the number of pipe legs meeting at a point does not change the symbol — an elbow with two marks is two connections, a tee with three marks is three connections.